



香港科技大學
THE HONG KONG
UNIVERSITY OF SCIENCE
AND TECHNOLOGY

COMP 5212

Machine Learning

Instructor: Zihan Zhang

Course website: <https://zsubfuncz.github.io/COMP5212-2026Fall/>

Canvas code: 72122

Pre-requisite

- Probability
 - Distribution, random variable, expectation, conditional distribution, variance
- Linear algebra
 - Matrix multiplication
- Python programming
 - Numpy, torch

Grading

- 5 assignments (40%)
 - 3 Written assignments (20%) + 2 Programming assignments (20%)
 - 3 free late days in total, for additional late days, 20% penalization applied for each day late
 - No assignment will be accepted more than 3 days late
- Mid-term Exam, open-note (25%)
- Final exam, open-note (35%)

Honor Code

Do's

- Form study groups to discuss (e.g. homework)
- Write down the homework solutions independently
- Write down the names of people with whom you've discussed the homework
- You are encouraged to use generative AI (e.g. ChatGPT) to assist you

Don'ts

- Copy, refer to, or look at solutions from previous years, online, or others
- Copy ChatGPT's answers
- Longer versions on the course website

We have zero tolerance — in the case of honor code violation for a single time, you will fail this course directly

Waiting List & Audit

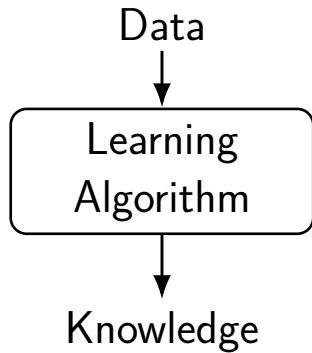
- We will try our best to include more students on the waiting list.
- Audit is allowed, and will be graded in the same criterion

More Info on Course Website

- Canvas is the main platform for announcement, discussion and homework submission
- Recorded videos on canvas
- Syllabus, slides and relevant reading materials
- Detailed course logistics

Overview of Machine Learning

What is Machine Learning



Machine Learning is Trending

- Everywhere — wide application
 - Finance, data scientist, medical diagnosis, translation, self-driving....
- Foundation of artificial intelligence — one of the most important technology for the society in the next 10s of years
 - ChatGPT, large language model, large multimodal model

Machine Learning Paradigms

- Supervised learning
- Unsupervised learning
- Self-supervised learning
- Reinforcement learning

Supervised Learning

Housing Price Prediction

- Given: a dataset that contains n samples

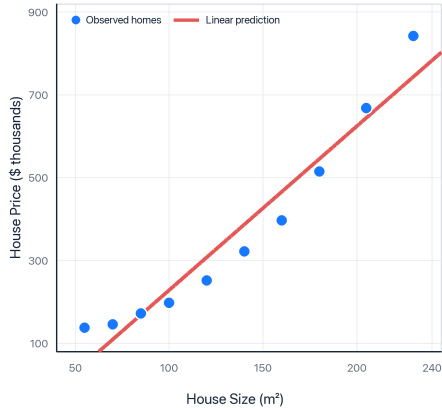
$$(x^{(1)}, y^{(1)}), \dots, (x^{(n)}, y^{(n)})$$

- Task: if a residence has x square feet, predict its price?

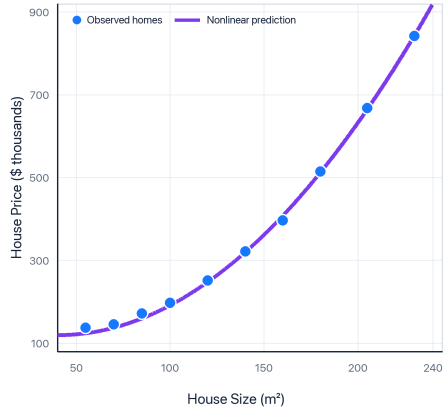
Example from Stanford CS229

Housing Price Prediction

Linear Price Prediction



Nonlinear Price Prediction



More Features

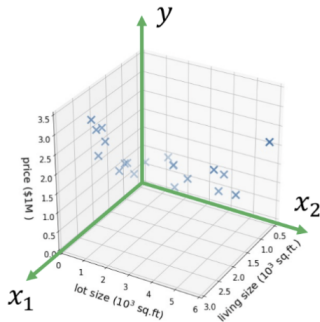
- Suppose we also know the lot size

Example from Stanford CS229

More Features

- Suppose we also know the lot size
- Task: find a function that maps

$$\underbrace{(\text{size, lot size})}_{\substack{\text{features/input} \\ x \in \mathbb{R}^2}} \longrightarrow \underbrace{\text{price}}_{\substack{\text{label/output} \\ y \in \mathbb{R}}}$$



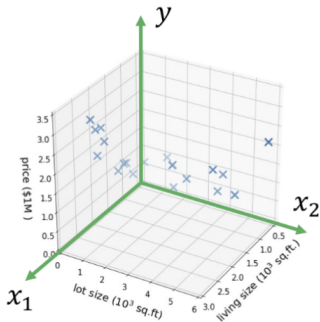
More Features

- Suppose we also know the lot size
- Task: find a function that maps

$$\underbrace{(\text{size, lot size})}_{\substack{\text{features/input} \\ x \in \mathbb{R}^2}} \longrightarrow \underbrace{\text{price}}_{\substack{\text{label/output} \\ y \in \mathbb{R}}}$$

- Dataset: $(x^{(1)}, y^{(1)}), \dots, (x^{(n)}, y^{(n)})$

where $x^{(i)} = (x_1^{(i)}, x_2^{(i)})$

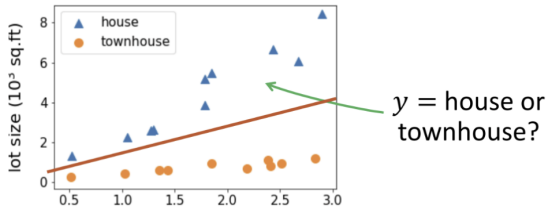


High-dimensional Features

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ \vdots \\ x_d \end{bmatrix} \quad \begin{array}{l} \text{— living size} \\ \text{— lot size} \\ \text{— \# floors} \\ \text{— condition} \\ \text{— zip code} \\ \vdots \end{array} \quad \longrightarrow \quad y \text{ — price}$$

Regression vs Classification

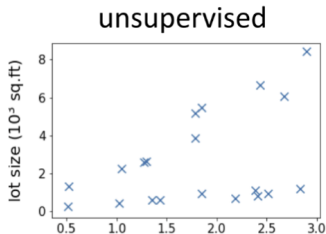
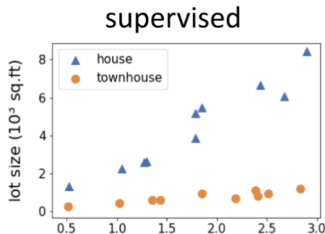
- Regression: if $y \in \mathbb{R}$ is a continuous variable
 - E.g., price prediction
- Classification: the label is a discrete variable
 - E.g., predicting the types of residence



Unsupervised Learning

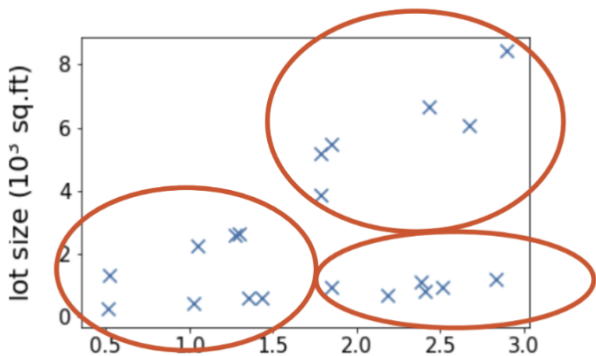
Unsupervised Learning

- The dataset contains no labels: $x^{(1)}, \dots, x^{(n)}$
- The goal is typically vague: find interesting structures in the data

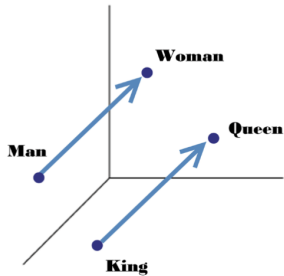


Examples from CS229

Clustering

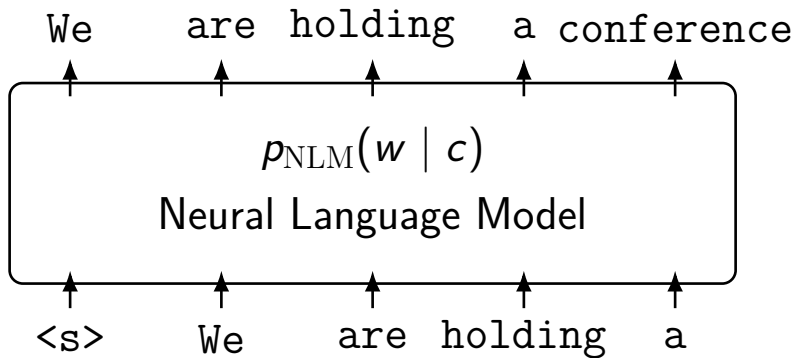


Word Embeddings



Self-supervised Learning

Language Models



The label is from the data itself.

Large Language Models

An autoregressive model predicts each token using only the tokens that appear before it:

$$P(x_1, \dots, x_T) = \prod_{t=1}^T P(x_t \mid x_1, \dots, x_{t-1}).$$

Training examples

Input context

The

The cat

The cat sat

The cat sat on

The cat sat on the

Target

cat

sat

on

the

mat

Generation

Machine

→ Machine learning

→ Machine learning can

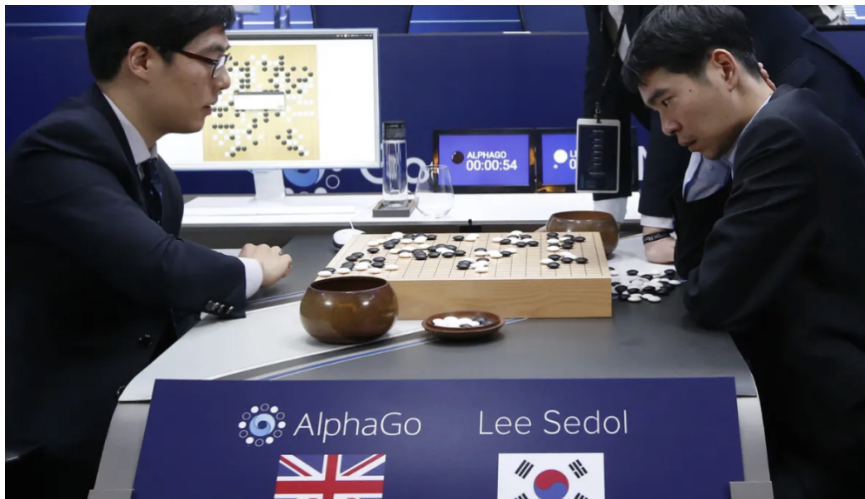
→ Machine learning can

predict

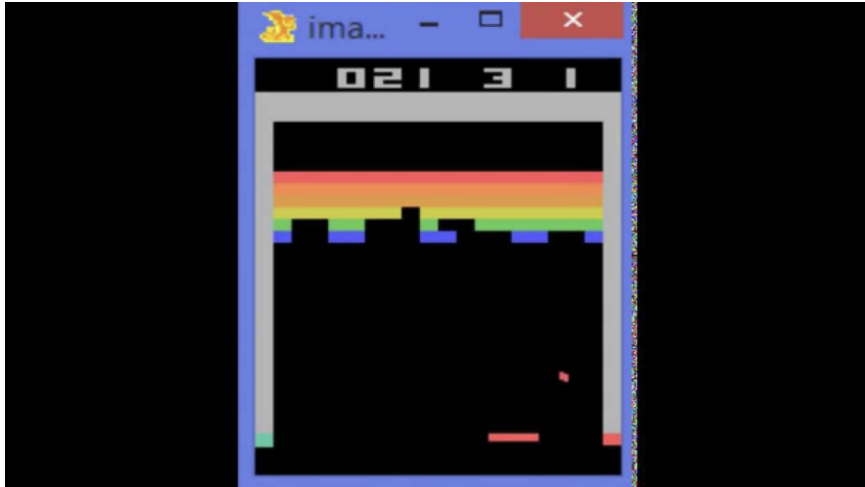
→ ...

Reinforcement Learning

AlphaGo

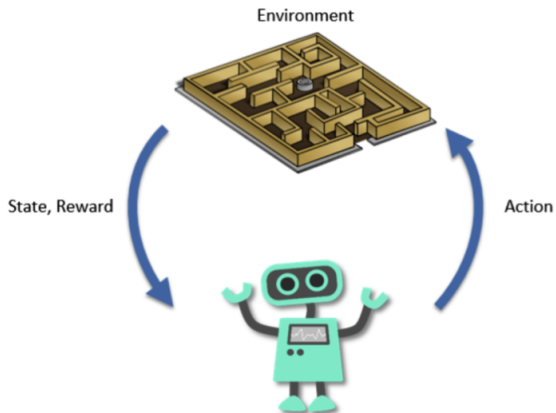


Atari Breakout Game



Reinforcement Learning

- RL can collect data **interactively**



Overview of Topics

Introduction
Math basics
Foundations of machine learning
Linear/Logistic regression
Generalized linear models, classification
Kernel methods
SVM
MLE
Gradient descent, SGD, Newton's method
Generalization, bias-variance tradeoff
Clustering
Expectation Maximization
PCA/ICA
mid-term exam

Basics of FNN
Neural networks architectures (CNN/RNN)
Variational autoencoder
Generative adversarial networks
Stable diffusion
Reinforcement Learning
Large language models

Thank You!

Questions?